

ManuTrace AI™ — Exhaustive Feature Specification

Enterprise Industrial AI, Digital Product Passport (DPP) & Supply Chain Traceability Operating System

Document Version: 2.4 LTS | **System Reference:** MT-SPEC-2026-V2

Executive Summary: ManuTrace AI is an enterprise-grade Industrial AI platform achieving zero-defect manufacturing, real-time supply chain transparency, and automated root cause diagnostics across 5 industrial sectors. It provides cryptographic forward genealogy, backward blast-radius containment, 10 Hz live machine telemetry streaming, multi-variate statistical process control (SPC) anomaly detection, and a digital twin process simulator.



1. Platform Architecture & Infrastructure

Layer	Technologies / Standard	Operational Role
Frontend Framework	React 18.3 (TypeScript), Vite 6.4, Tailwind CSS	Ultra-responsive client-side web application with BankDash modern design aesthetics.
Intelligence Engine	Multi-variate SPC Anomaly Detector, Regression RCA Engine, Digital Twin What-If Simulator	Autonomous heuristics calculating real-time statistical deviations and physical correlations.
Audio Synthesizer	Native Web Audio API Industrial UI Synthesizer	Zero-asset audio synthesis generating distinct wave frequencies for clicks, verifications, alerts, and scans.
Compliance Standards	ISO 9001:2015, IATF 16949, AS9100D, cGMP (FDA 21 CFR Part 210), EU Digital Product Passport	Standardized regulatory schemas with cryptographic SHA-256 integrity anchoring.



2. Multi-Sector Configurable Engine (5 Industrial Sectors)

- 🚗 **Automotive & Precision Engineering:** Focus on brake calipers, transmission housings, CNC tolerances, and IATF 16949 compliance.
- ⚡ **Semiconductor & Electronics:** Focus on silicon wafers, GaN substrates, SMT lines, and Sony Green Partner certification.
- 💊 **Pharma & BioTech (cGMP):** Focus on active pharmaceutical ingredients (API), lyophilized vials, and FDA 21 CFR Part 210 compliance.
- 🍷 **Food & Beverage Processing:** Focus on certified organic concentrates, pasteurization thermal envelopes, and HACCP / FSMA standards.
- 🧵 **Technical Textiles & High-Performance Apparel:** Focus on high-tenacity yarn, tensile strength, and OEKO-TEX Standard 100.



3. Master UI Layout, Navigation & Productivity Hubs

3.1 Collapsible Left Sidebar

- **Brand Header:** ManuTrace AI logo with dynamic sector badge, factory icon, and collapse/expand toggle button.
- **16 Navigation Modules:** Each item features custom icon, active blue vertical indicator bar, and color-coded status badges (CORE, LOAD, KEY, AI, NEW, Alert).
- **Collapsed Mode:** Minimizes to icon rail (w-20) with hover tooltips and expand button.
- **Live Telemetry Status Box:** Shows real-time 10 Hz synchronization status, pulsing green beacon, active build version (v2.4 LTS), and audio synthesizer mute/unmute toggle.

3.2 Top Header & Contextual Ribbon Bar

- **Universal Search Bar:** Clickable search button with search icon, placeholder text, and Ctrl K badge.
- **AI Copilot Quick Button:** Integrated button launching the natural language assistant.
- **Mission Tagline Banner:** "TRACE EVERYTHING. UNDERSTAND ANYTHING. ACT BEFORE IMPACT."
- **10 Hz Telemetry Toggle:** Displays live stream pulse (10 Hz Live vs Paused) with toggle sound.
- **1-Click Scenarios Button:** Amber pill opening the preset test scenario launcher.
- **5-Min Expo Tour Button:** Gradient button with spinning sparkles icon launching the 17-step guided story.

- **Industrial Sector Dropdown:** Dynamic sector selector with emoji icons (🚚 , ⚡ , 💊 , 🇺🇸 , 🏭) switching data schemas live.
- **Notification Bell:** Circular button with unread counter and pulsing red indicator for critical alerts.
- **Contextual Ribbon Bar (Below Header):** Shows module breadcrumbs, in-place Batch Selector dropdown, in-place Machine Selector dropdown, and speed-dial action pills (RCA, Reverse Trace, Passport, Simulator, Scenarios, Audit PDF, Ask AI).

3.3 Global Modals & Productivity Drawers

- 🗨️ **ManuTrace AI Copilot Modal (`Ctrl + J`):** Context-aware natural language assistant answering any question with live data, metric cards, simulated voice input, and 1-click action buttons.
- 🎯 **1-Click Scenario Matrix Modal:** 4 instant test scenarios (Critical Thermal Drift RCA, Raw Material Variance Trace, Spindle Vibration Twin Simulator, Zero-Defect EU DPP).
- 📄 **Printable Executive Audit Dossier Modal (`Ctrl + E`):** Formats complete ISO 9001 / EU DPP compliance report with print stylesheet and JSON copy.
- 📱 **Digital Passport QR Modal:** Public tamper-evident QR code and unmasked internal auditor view toggle.
- 🔍 **Universal Global Search Modal (`Ctrl + K`):** Instant modal search across batches, products, machines, suppliers, raw materials, and shipments.
- 🚀 **5-Minute Guided Expo Story Tour (`Ctrl + D`):** 17-step guided narrative walkthrough of the full platform storyline.
- 📖 **Keyboard Shortcuts Guide (`?`):** Visual modal displaying all productivity hotkeys.
- 🔔 **Real-Time Action Toasts:** Instant visual feedback for all user operations.

🚀 4. Exhaustive Specification of All 17 Core Modules

Module 1: Overview Dashboard (/dashboard)

- **Hero Passport Cards:** Primary Tracked Lot (Batch B-1042 with Anomaly Flagged badge) and Certified Benchmark Lot (Batch B-1041 with Pass 99.4% FPY badge).
- **Recent Incident Stream:** Real-time log of thermal spikes, vibration anomalies, and material hardness variations with timestamps.

- **KPI Quad:** Active Lots (24 Batches), Units Produced (18,426), First-Pass Yield (96.8%), and Quarantined Lots (4 Batches).
- **Weekly Production Double-Bar Chart:** Visual comparison of target units vs completed units across days with Thursday spike callout.
- **Manufacturing Health Score Donut (87/100):** Process Health (92%), Quality (96.8%), Supply (82%), Machine (78%).
- **1-Click Scenario Matrix:** 4 cards launching pre-configured test scenarios.
- **Quick Containment Card:** 1-click button to contain Batch B-1042 blast radius and launch rework simulation.

Module 2: Industrial Dataset Studio (/dataset)

- **Universal Drag & Drop Ingestion:** Upload CSV, JSON, or Excel files for Batches, Products, Machines, Suppliers, Raw Materials, Inspections, Defects, and Shipments.
- **Dual Ingestion Modes:** Append Mode (adds new rows to memory) vs Replace Mode (replaces database).
- **Preloaded Sector Benchmarks:** 1-Click loading of industry datasets for all 5 sectors.
- **Active Database Explorer:** Live searchable table displaying all rows in memory with column sorting, record count, and risk breakdown.
- **Dataset Export & Reset:** Export active datasets to JSON/CSV or revert to baseline factory defaults.

Module 3: Digital Product Passport (/passport)

- **7-Stage Lifecycle Verification Chain:**
 1. *Raw Material Reception:* Lot purity score, mill test certificate, received date.
 2. *Supplier Verification:* Tier-1 quality rating, ISO/IATF credentials.
 3. *Machine Processing Cell:* Chamber temperature, hydraulic pressure, spindle vibration, operator ID.
 4. *Assembly & Production:* Cycle dwell time, serialization, line assignment.
 5. *Quality Inspection Gateway:* Eddy current testing, surface roughness, pass/fail disposition.
 6. *Packaging & QR Serialization:* Cryptographic SHA-256 seal, blockchain anchor.
 7. *Logistics & Distribution:* Carrier tracking, customer dispatch, dock quarantine status.
- **Cryptographic Verification Seal:** SHA-256 digital signature guaranteeing authenticity.

- **Dual View Modes:** Public Consumer View (sanitized) vs Certified Internal Auditor View (unmasked raw telemetry).
- **Interactive Digital Twin Visualizer:** 3D-styled component breakdown showing internal sensor telemetry.

Module 4: Products & Batches Directory (/products)

- Searchable master catalog of all manufactured units and active lots.
- Multi-factor filters by Sector, Quality Status, Risk Level, and Production Line.
- Direct 1-click links to RCA, Passport, Simulator, and QR modal.

Module 5: Forward Traceability Flow (/traceability)

- Multi-tier genealogical node graph from raw material supplier to customer delivery.
- Node-by-node inspection of timestamps, operator badges, recorded temperatures, and pass rates.

Module 6: Backward / Reverse Traceability & Blast Radius (/reverse-trace)

- Upstream root tracing from any raw material lot (e.g. RM-7821) back to point of origin.
- Automated Blast Radius Calculation: 3 compromised batches, 486 finished units, 7 quarantined customer shipments.
- 1-Click Precision Containment: Simultaneous quarantine of all affected downstream entities.

Module 7: AI Anomaly Intelligence (/ai-intelligence)

- Real-time statistical process control (SPC) anomaly detector calculating Z-score deviations against 3-sigma limits.
- Multi-variate risk rating (Thermal 40%, Vibration 25%, Supplier 20%, Process 15%).
- Severity categorization (Low, Medium, High, Critical) with recommended engineering actions.

Module 8: Root Cause Analysis (/rca)

- Contributing Factors Tree: 42% Temperature Deviation, 24% Spindle Vibration, 18% Material Hardness, 11% Processing Time, 5% Operator Clamping Torque.
- Direct sensor evidence correlation proving root cause.
- Corrective and Preventive Action (CAPA) generator.

Module 9: Impact Analysis & Risk Forensics (/impact)

- Cascading operational risk evaluation and financial exposure calculator (\$42,500 USD).
- Multi-tier impact tree tracking units at Triggering Event, Active Lots, Finished Assemblies, QA Gates, and Logistics.

Module 10: What-If Digital Twin Process Simulator (/simulator)

- Real-time sliders for Chamber Temperature, Hydraulic Pressure, Processing Time, and Spindle Speed.
- Presets: Nominal Standard, High Speed / Throughput, Eco / Energy Saver, Stress Test / Thermal Limit.
- Automated Stress Test Run: Dynamically ramps temperature by +3°C every 250ms with telemetry audio chirps to locate failure threshold.
- Real-time predictive comparison between Current Baseline vs Simulated Process across Yield %, Defect Risk %, Energy (kWh), and Tooling Stress.

Module 11: Quality Control & Inspection Gateways (/quality)

- Station-by-station inspection logs (Eddy Current, Surface Finish, CMM Dimensional Tolerances).
- Defect Category Pareto breakdown (Dimension, Surface, Material, Assembly, Packaging).

Module 12: Supplier Scorecard & Performance Matrix (/suppliers)

- Tier-1 supplier leaderboard, quality scores, defect rates, and active batch counts.
- Compliance certificates (IATF 16949, ISO 9001:2015, AS9100D, cGMP, HACCP).

Module 13: Machine Fleet & Spindle Telemetry (/machines)

- Fleet telemetry monitor for Machines M01 to M06.
- Spindle harmonic vibration FFT / RMS logs against ISO 10816 limits (< 2.8 mm/s nominal).
- 10 Hz Live Stream Simulation toggle with dynamic industrial jitter.

Module 14: Executive Operational Analytics (/analytics)

- Overall Equipment Effectiveness (OEE) analytics (92.4% Fleet OEE).
- First-Pass Yield (FPY) trend curves and Cost of Quality (CoQ) breakdown.

Module 15: Compliance & Audit Reports (/reports)

- 1-Click formal ISO 9001:2015 compliance audit package generator.
- EU Digital Product Passport JSON-LD & PDF exports.

Module 16: System Settings & Governance (/settings)

- Dynamic industrial sector switching and user role administration.
- LocalStorage backup and factory default restore.

Module 17: Enterprise Authentication (/login)

- Role-based Single Sign-On simulation (Director, QA Lead, Process Engineer, Auditor).
- 1-Click fast demo access for all 5 sectors.

 5. Mathematical SPC Formulas & Empirical Models

SPC Z-Score Anomaly Formula:
 $Z = |T_{\text{current}} - T_{\text{baseline}}| / \sigma$
(Where $T_{\text{baseline}} = 180.0^{\circ}\text{C}$ and $\sigma = 2.5^{\circ}\text{C}$. Anomaly triggered when $Z > 2.0$ or $|T - 180| > 5^{\circ}\text{C}$)

Multi-Variate Composite Risk Rating:
 $\text{Risk}_{\text{composite}} = 0.40 \times R_{\text{thermal}} + 0.25 \times R_{\text{vibration}} + 0.20 \times R_{\text{supplier}} + 0.15 \times R_{\text{variance}}$

What-If Process Simulation Response Functions:
 $\Delta\text{QualityRisk} = 2.4 \times \Delta T + 3.1 \times \Delta P - 0.8 \times \Delta t$
 $\Delta\text{DefectRisk} = 1.6 \times \Delta T + 1.2 \times (\Delta\text{Speed} / 100) + 1.8 \times \Delta P$
 $\Delta\text{Energy} = 1.0 \times \Delta T + 2.5 \times (\Delta\text{Speed} / 100) + 1.5 \times \Delta t$
 $\text{Yield}_{\text{sim}} = \max(60, \min(99.8, \text{Yield}_{\text{current}} - 0.4 \times \Delta\text{DefectRisk}))$

 6. Web Audio Synthesizer Sound Design

Sound Action	Waveform	Frequency Ramp	Gain & Duration
Button Click Tick	Sine Wave	800 Hz → 400 Hz (exponential)	0.04 Gain, 40 ms duration

Sound Action	Waveform	Frequency Ramp	Gain & Duration
Verification Chime	Dual Sine Chord	D5 (587.3 Hz) / A5 (880 Hz) → A5 / D6 (1174.6 Hz)	0.05 Gain, 350 ms duration
Alert Warning Buzz	Sawtooth Wave	440 Hz → 370 Hz	0.06 Gain, 220 ms duration
QR Laser Scanner	Triangle Wave	1200 Hz → 2400 Hz	0.06 Gain, 120 ms duration
Quarantine Sub-Bass	Deep Triangle	160 Hz → 45 Hz (sub-bass drop)	0.12 Gain, 350 ms duration
Telemetry Packet Tick	Sine Wave	1800 Hz	0.015 Gain, 20 ms duration

7. Keyboard Shortcuts & Productivity Hotkeys

Shortcut	Action Description
Ctrl + K / Cmd + K	Open Universal Global Search Modal (Search Lot, Machine, Supplier, Material, Shipment)
Ctrl + J / Ctrl + /	Launch ManuTrace AI Copilot Assistant
Ctrl + D	Start / Exit 5-Minute Guided Expo Story Tour (17 Steps)
Ctrl + E	Generate & Print ISO 9001 Batch Audit Dossier
? (Shift + /)	Open Keyboard Shortcuts & Productivity Guide
ESC	Close active modal, drawer, or search overlay